

The Orogen Project

Open resource of geoscience exploration

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The large language models that define modern artificial intelligence became possible because the world's libraries were digitized. Decades of patient, deliberate work to scan, tokenize, and federate the accumulated text of human civilization produced a corpus large enough that, once exposed to it, a model could learn to read, synthesize and generate text.

The Earth's subsurface is waiting for the same moment. Geological maps, borehole logs, geophysical surveys, and geochemical analyses representing more than a century of public investment in understanding our home planet reside in scanned documents, institutional repositories, and disciplinary silos, unavailable to modern AI systems. A common saying in geology is that the best geologists are those who have seen the most rocks. The same is true of predictive models. The data already exists. What is missing is the infrastructure to utilize it.

The Open Resource of Geoscience Exploration (Orogen) Project is building that infrastructure. The technical term *orogen* refers to a mountain belt in which deep crustal rocks have been exhumed and exposed at the surface. Our goal is to expose the buried archive of geoscience knowledge and make it findable, accessible, interoperable, and reusable. By doing so, we can teach a machine how to interpret observations of the Earth.

What are you trying to do?

We are building the open digital substrate of the Earth's upper crust. Concretely, we aim to extract the contents of the public geoscience archives (geologic maps, borehole logs, geophysical surveys, and geochemical analyses) from the legacy formats in which they are trapped, to federate them through a common schema so that data from different disciplines and sources can be synthesized, and to train physics-constrained foundation models that can predict what lies beneath any patch of the Earth's surface, with calibrated uncertainty. The result is a common baseline resource models can use.

How is it done today?

Today, every team that wants to ask a question of the subsurface starts from scratch. A geologist assembles a regional dataset from a handful of sources, fits a model and rarely publishes the data compilation. Disciplinary databases sit beside each other but are not designed to interoperate. AI prospectivity models are trained almost exclusively on known deposit locations, so they learn where things are, not where they are not. The federal and state archives, representing trillions of dollars in cumulative investment, are overwhelmingly inaccessible to modern models; they consist of scanned reports, paper maps, inconsistent coordinates – formats that were never designed to be machine-readable. We have directly sampled less than *one millionth* of the Earth's crust, and even so, we are underutilizing that information. The bottleneck is not the amount of data available, but rather the cost to access it.

What is new, and why will it work?

Three things are new, and all three converged only in the last two or three years.

First, the AI to extract the archive now exists. Document understanding, multimodal extraction, and geospatial parsing at production scale did not exist when this problem was first scoped. It does now. In the DARPA-funded [CriticalMAAS](#) program, which we led, we proved end-to-end that a computer can be taught to interpret a geologic [map](#) and reduce the time needed to conduct mineral resource assessment from years to [days](#). Orogen extends that capability across every major modality of subsurface data.

Second, the modeling paradigm fits the problem. Foundation models for the subsurface are physics-constrained x/y/z embeddings that encode Tobler's First Law of Geography: things that are close together are more related than things that are far apart. Using the same principles that allow language models predict the next word in a sentence, we tokenize the Earth's crust and learn how one crustal domain relates to another, producing predictions with calibrated uncertainty rather than confident guesses. Geological reasoning improves AI predictions with better attribution.

Third, the precedent exists. The Human Genome Project showed what happens when a foundational reference is treated as a public good. The digitization of libraries showed what happens when an archive becomes machine-readable. Geoscience has not yet had its analogous moment; Orogen is built to enable it.

Who cares, and what difference does it make?

The upper few kilometers of the Earth's crust is the reservoir for nearly everything modern society needs over the next century, and almost every one of those needs is constrained today by the same lack of subsurface understanding.

Critical minerals. The lithium, cobalt, nickel, copper, rare earths, tungsten, and other commodities required for the energy transition, advanced manufacturing, and defense all come from the crust. There is no substitute. A foundation model that can predict prospectivity from the subsurface archive directly accelerates discovery and reduces dependence on supply chains controlled by other nations.

Freshwater and aquifers. The aquifers that supply cities and agriculture are subsurface features whose boundaries, recharge rates, and storage capacities are still poorly characterized in many of the regions that depend on them most. The same subsurface models that find minerals can be used to map groundwater resources.

Geothermal energy. Enhanced geothermal systems are bottlenecked on the ability to predict subsurface temperature, permeability, and structure. A federated, AI-ready substrate is the missing input.

Hazard assessment. Earthquakes, landslides, sinkholes, and induced seismicity all originate in features the subsurface archive describes but no single agency can synthesize. A common substrate makes synthesis possible.

Public-good economics. Geoscience Australia has demonstrated a 500-to-1 [return](#) on investment for precompetitive, open-access geoscience data. The data already on the shelf in the United States is, in effect, a stranded asset measured in the trillions. Orogen is the

mechanism to unlock it. Better predictive models of the subsurface result in better definition of in situ resources. This information, in turn, helps reduce uncertainty and risk, contributing to wise resource stewardship.

Why now

Three things have converged for the first time, and the window is shorter than it looks.

The technology is new. Multimodal foundation models, large-scale document understanding, and geospatial embedding methods did not exist three years ago. They do now, and they are mature enough to deploy against the legacy archive at production scale.

The mandate is acute. Critical mineral supply security, the energy transition, water stress, and climate resilience are not abstract policy concerns. They are line items in federal budgets and state strategic plans, and each of them depends on a level of subsurface understanding that does not yet exist.

The archive is perishable. The records Orogen is built to ingest are physically deteriorating — paper maps, scanned PDFs of variable quality, magnetic tapes whose readers are out of production. The geologists who know how to interpret them are retiring. Every year that passes, a portion of the archive becomes irretrievable. This is not a problem that gets cheaper to solve later.

Why us

Jataware is a frontier research contractor (FRC) focused on the hardest challenges in national security and critical infrastructure. The capable team of software developers, engineers, and scientists with domain experience and a record of performance in ARPA-style programs is well suited to execute Orogen. By partnering with government agencies, national labs, universities, and other stakeholders, we can accelerate the work.

The call

The tools exist. The data exists. What is missing is the infrastructure to bring them together — the patient, deliberate work of digitizing, federating, and curating an archive built up over a century of public investment. That is the work Orogen is built to do. From our previous success with CriticalMAAS, we have proven we can teach a computer to read a map. Our ambition now is larger: to teach the machine to read the Earth.

With funding of \$5 million over 3 years, the next generation of discoveries in minerals, water, energy, and hazards is no longer a matter of who happens to hold the right file. It becomes a matter of asking the right question of an Earth we finally understand.

The Orogen Project

Curate the underground truth.

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